PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	29 October 2026	Maximum Marks	2 Marks
Team ID	EV-BI-04	Project Name	Electric Vehicle Market Performance Insights Dashboard

Define Your Technology Stack:
Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

Technology Stack Breakdown

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	Python, HTML, CSS, JS	Develop Flask web application backend and responsive frontend UI.
2.	**Database / Data Warehouse**	Warehouse used to store and manage structured business data for analysis.	Microsoft Excel (CSV)	Store EV market dataset cleanly for analysis and seamless integration.
3.	**Data Ingestion**	Used to load data from source files/systems into analytical environment.	Tableau Data Connection	Import raw dataset efficiently into Tableau analytical environment.
4.	**Data Transformation & Logic**	Tools/logic used to clean, join, filter, and calculate analytical KPIs.	Tableau Desktop	Clean, filter, transform EV data, and calculate market metrics & rules.
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, and visual analysis.	Tableau Public	Build interactive dashboards, KPI charts, and visual market stories.
6.	**Front-End / User Interface**	Interface through which users interact with dashboards and filters.	Flask + HTML + CSS + JS	User-friendly responsive website integrating embedded dashboards.
7.	**Version Control**	Track code history and facilitate seamless team collaboration.	Git & GitHub	Project management, source code versioning, and repository hosting.
8.	**Deployment & Hosting**	Publish visual reports and hosting platforms for client usage.	Tableau Public + Flask	Publish live interactive dashboards and host the web application.